In summary, the problem of FIR filter design is simply to determine the M coefficients $h(n)$, $n = 0, 1, \ldots, M-1$, from a specification of the desired frequency response $H_d(\omega)$ of the FIR filter. The important parameters in the specification of $H_d(\omega)$ are given in Fig. 10.1.2.

In the following subsections we describe design methods based on specification of $H_d(\omega)$.

10.2.2 Design of Linear-Phase FIR Filters Using Windows

In this method we begin with the desired frequency response specification $H_d(\omega)$ and determine the corresponding unit sample response $h_d(n)$. Indeed, $h_d(n)$ is related to $H_d(\omega)$ by the Fourier transform relation

$$H_d(\omega) = \sum_{n=0}^{\infty} h_d(n) e^{-j\omega n} \tag{10.2.17}$$

where

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega \tag{10.2.18}$$

Thus, given $H_d(\omega)$, we can determine the unit sample response $h_d(n)$ by evaluating the integral in (10.2.17).

In general, the unit sample response $h_d(n)$ obtained from (10.2.17) is infinite in duration and must be truncated at some point, say at $n = M-1$, to yield an FIR filter of length M. Truncation of $h_d(n)$ to a length $M-1$ is equivalent to multiplying $h_d(n)$ by a "rectangular window," defined as

$$w(n) = \begin{cases} 1, & n = 0, 1, \ldots, M-1 \\ 0, & \text{otherwise} \end{cases} \tag{10.2.19}$$

Thus the unit sample response of the FIR filter becomes

$$\begin{aligned} h(n) &= h_d(n) w(n) \\ &= \begin{cases} h_d(n), & n = 0, 1, \ldots, M-1 \\ 0, & \text{otherwise} \end{cases} \end{aligned} \tag{10.2.20}$$

It is instructive to consider the effect of the window function on the desired frequency response $H_d(\omega)$. Recall that multiplication of the window function $w(n)$ with $h_d(n)$ is equivalent to convolution of $H_d(\omega)$ with $W(\omega)$, where $W(\omega)$ is the frequency-domain representation (Fourier transform) of the window function, that is,

$$W(\omega) = \sum_{n=0}^{M-1} w(n) e^{-j\omega n} \tag{10.2.21}$$

Thus the convolution of $H_d(\omega)$ with $W(\omega)$ yields the frequency response of the (truncated) FIR filter. That is,

$$H(\omega) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(v)W(\omega - v)dv \tag{10.2.22}$$

The Fourier transform of the rectangular window is

$$\begin{aligned} W(\omega) &= \sum_{n=0}^{M-1} e^{-j\omega n} \\ &= \frac{1 - e^{-j\omega M}}{1 - e^{-j\omega}} = e^{-j\omega(M-1)/2}\,\frac{\sin(\omega M/2)}{\sin(\omega/2)} \end{aligned} \tag{10.2.23}$$

This window function has a magnitude response

$$|W(\omega)| = \frac{|\sin(\omega M/2)|}{|\sin(\omega/2)|}, \qquad -\pi \le \omega \le \pi \tag{10.2.24}$$

and a piecewise linear phase

$$\Theta(\omega) = \begin{cases} -\omega\left(\dfrac{M-1}{2}\right), & \text{when } \sin(\omega M/2) \ge 0 \\ -\omega\left(\dfrac{M-1}{2}\right) + \pi, & \text{when } \sin(\omega M/2) < 0 \end{cases} \tag{10.2.25}$$

The magnitude response of the window function is illustrated in Fig. 10.2.2 for $M = 31$ and 61. The width of the main lobe [width is measured to the first zero of $W(\omega)$] is $4\pi/M$. Hence, as M increases, the main lobe becomes narrower. However, the sidelobes of $|W(\omega)|$ are relatively high and remain unaffected by an increase in M. In fact, even though the width of each sidelobe decreases with an increase in M, the height of each sidelobe increases with an increase in M in such a manner that the area under each sidelobe remains invariant to changes in M. This characteristic behavior is not evident from observation of Fig. 10.2.2 because $W(\omega)$ has been normalized by M such that the normalized peak values of the sidelobes remain invariant to an increase in M.

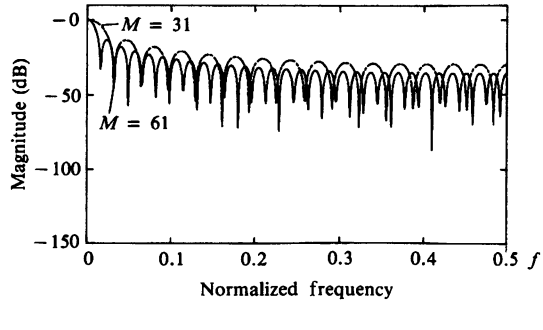


Figure 10.2.2
Frequency response for rectangular window of lengths (a) $M = 31$, (b) $M = 61$.

The characteristics of the rectangular window play a significant role in determining the resulting frequency response of the FIR filter obtained by truncating $h_d(n)$ to length M. Specifically, the convolution of $H_d(\omega)$ with $W(\omega)$ has the effect of smoothing $H_d(\omega)$. As M is increased, $W(\omega)$ becomes narrower, and the smoothing provided by $W(\omega)$ is reduced. On the other hand, the large sidelobes of $W(\omega)$ result in some undesirable ringing effects in the FIR filter frequency response $H(\omega)$, and also in relatively larger sidelobes in $H(\omega)$. These undesirable effects are best alleviated by the use of windows that do not contain abrupt discontinuities in their time-domain characteristics, and have correspondingly low sidelobes in their frequency-domain characteristics.

Table 10.1 lists several window functions that possess desirable frequency response characteristics. Figure 10.2.3 illustrates the time-domain characteristics of the windows. The frequency response characteristics of the Hanning, Hamming, and Blackman windows are illustrated in Figs. 10.2.4 through 10.2.6. All of these window

TABLE 10.1 Window Functions for FIR Filter Design

Name of window	Time-domain sequence, $h(n), 0 \le n \le M-1$
Bartlett (triangular)	$1 - \dfrac{2\left\lvert n - \dfrac{M-1}{2}\right\rvert}{M-1}$
Blackman	$0.42 - 0.5\cos\dfrac{2\pi n}{M-1} + 0.08\cos\dfrac{4\pi n}{M-1}$
Hamming	$0.54 - 0.46\cos\dfrac{2\pi n}{M-1}$
Hanning	$\dfrac{1}{2}\left(1 - \cos\dfrac{2\pi n}{M-1}\right)$
Kaiser	$\dfrac{I_0\left[\alpha\sqrt{\left(\dfrac{M-1}{2}\right)^2 - \left(n - \dfrac{M-1}{2}\right)^2}\right]}{I_0\left[\alpha\left(\dfrac{M-1}{2}\right)\right]}$
Lanczos	$\left\{\dfrac{\sin\left[2\pi\left(n - \dfrac{M-1}{2}\right)\Big/(M-1)\right]}{2\pi\left(n - \dfrac{M-1}{2}\right)\Big/\left(\dfrac{M-1}{2}\right)}\right\}^L, \quad L > 0$
	$1, \quad \left\lvert n - \dfrac{M-1}{2}\right\rvert \le \alpha\dfrac{M-1}{2}, \qquad 0 < \alpha < 1$
Tukey	$\dfrac{1}{2}\left[1 + \cos\left(\dfrac{n - (1+a)(M-1)/2}{(1-\alpha)(M-1)/2}\pi\right)\right]$
	$\alpha(M-1)/2 \le \left\lvert n - \dfrac{M-1}{2}\right\rvert \le \dfrac{M-1}{2}$

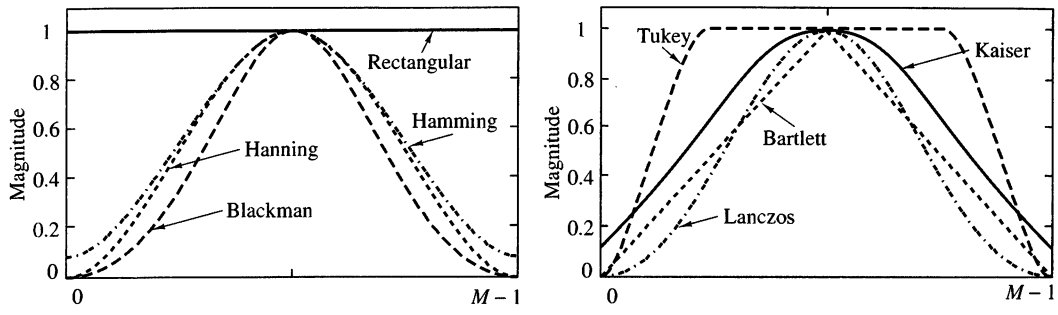


Figure 10.2.3 Shapes of several window functions.

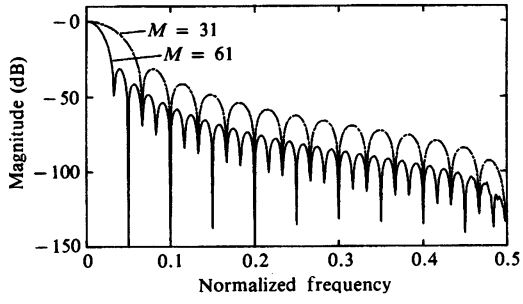


Figure 10.2.4
Frequency responses of Hanning window for (a) $M = 31$ and (b) $M = 61$.

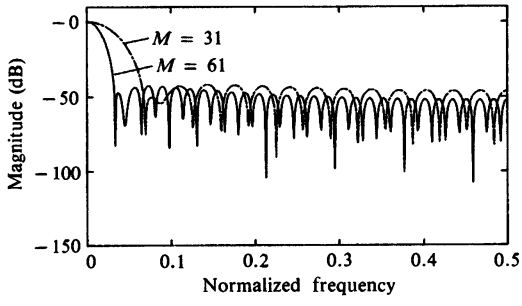


Figure 10.2.5
Frequency responses for Hamming window for (a) $M = 31$ and (b) $M = 61$.

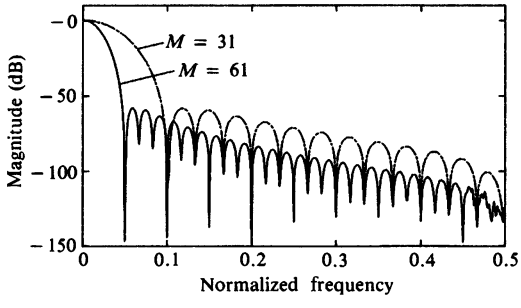


Figure 10.2.6
Frequency responses for Blackman window for (a) $M = 31$ and (b) $M = 61$.

functions have significantly lower sidelobes compared with the rectangular window. However, for the same value of M, the width of the main lobe is also wider for these windows compared to the rectangular window. Consequently, these window functions provide more smoothing through the convolution operation in the frequency domain, and as a result, the transition region in the FIR filter response is wider. To reduce the width of this transition region, we can simply increase the length of the window, which results in a larger filter. Table 10.2 summarizes these important frequency-domain features of the various window functions.

The window technique is best described in terms of a specific example. Suppose that we want to design a symmetric lowpass linear-phase FIR filter having a desired frequency response

$$H_d(\omega) = \begin{cases} 1e^{-j\omega(M-1)/2}, & 0 \le |\omega| \le \omega_c \\ 0, & \text{otherwise} \end{cases} \tag{10.2.26}$$

A delay of $(M-1)/2$ units is incorporated into $H_d(\omega)$ in anticipation of forcing the filter to be of length M. The corresponding unit sample response, obtained by evaluating the integral in (10.2.18), is

$$\begin{aligned} h_d(n) &= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega\left(n-\frac{M-1}{2}\right)}d\omega \\ &= \frac{\sin\omega_c\left(n-\dfrac{M-1}{2}\right)}{\pi\left(n-\dfrac{M-1}{2}\right)}, \qquad n \neq \frac{M-1}{2} \end{aligned} \tag{10.2.27}$$

Clearly, $h_d(n)$ is noncausal and infinite in duration.

If we multiply $h_d(n)$ by the rectangular window sequence in (10.2.19), we obtain

TABLE 10.2 Important Frequency-Domain Characteristics of Some Window Functions

Type of window	Approximate transition width of main lobe	Peak sidelobe (dB)
Rectangular	$4\pi/M$	−13
Bartlett	$8\pi/M$	−25
Hanning	$8\pi/M$	−31
Hamming	$8\pi/M$	−41
Blackman	$12\pi/M$	−57

an FIR filter of length M having the unit sample response

$$h(n) = \frac{\sin \omega_c \left(n - \dfrac{M-1}{2}\right)}{\pi \left(n - \dfrac{M-1}{2}\right)}, \qquad 0 \le n \le M-1, \quad n \ne \frac{M-1}{2} \tag{10.2.28}$$

If M is selected to be odd, the value of $h(n)$ at $n = (M-1)/2$ is

$$h\left(\frac{M-1}{2}\right) = \frac{\omega_c}{\pi} \tag{10.2.29}$$

The magnitude of the frequency response $H(\omega)$ of this filter is illustrated in Fig. 10.2.7 for $M = 61$ and $M = 101$. We observe that relatively large oscillations or ripples occur near the band edge of the filter. The oscillations increase in frequency as M increases, but they do not diminish in amplitude. As indicated previously, these large oscillations are the direct result of the large sidelobes existing in the frequency characteristic $W(\omega)$ of the rectangular window. As this window function is convolved with the desired frequency response characteristic $H_d(\omega)$, the oscillations occur as the large constant-area sidelobes of $W(\omega)$ move across the discontinuity that exists in $H_d(\omega)$. Since (10.2.17) is basically a Fourier series representation of $H_d(\omega)$, the multiplication of $h_d(n)$ with a rectangular window is identical to truncating the Fourier series representation of the desired filter characteristic $H_d(\omega)$. The truncation of the Fourier series is known to introduce ripples in the frequency response characteristic $H(\omega)$ due to the nonuniform convergence of the Fourier series at a discontinuity. The oscillatory behavior near the band edge of the filter is called the *Gibbs phenomenon*.

To alleviate the presence of large oscillations in both the passband and the stopband, we should use a window function that contains a taper and decays toward zero gradually, instead of abruptly, as it occurs in a rectangular window. Figures 10.2.8 through 10.2.11 illustrate the frequency response of the resulting filter when some of the window functions listed in Table 10.1 are used to taper $h_d(n)$. As illustrated in Figs. 10.2.8 through 10.2.11, the window functions do indeed eliminate the ringing effects at the band edge and do result in lower sidelobes at the expense of an increase in the width of the transition band of the filter.

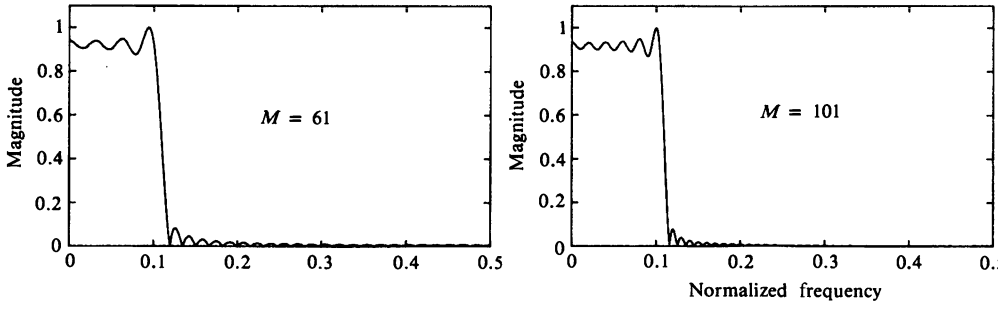


Figure 10.2.7 Lowpass filter designed with a rectangular window: (a) $M = 61$ and (b) $M = 101$.

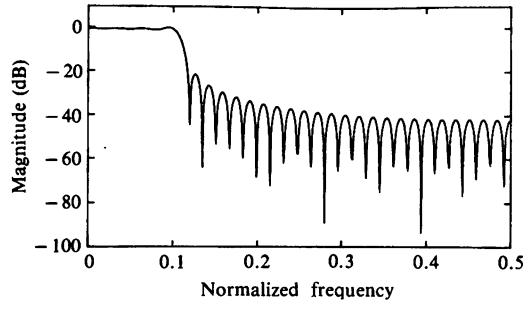


Figure 10.2.8
Lowpass FIR filter designed with rectangular window ($M = 61$).

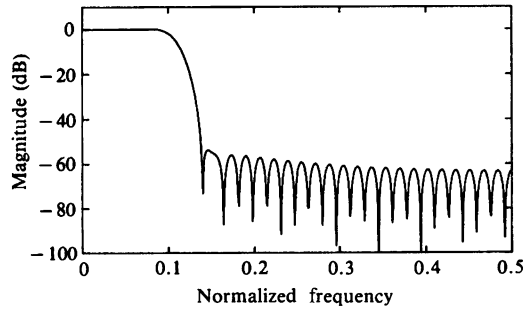


Figure 10.2.9
Lowpass FIR filter designed with Hamming window ($M = 61$).

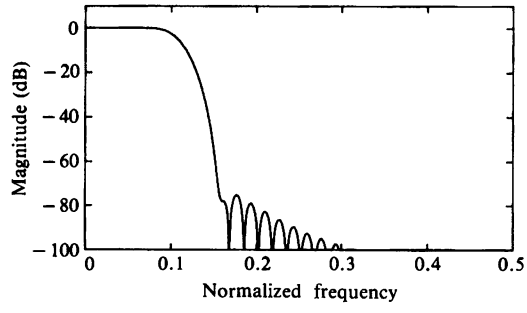


Figure 10.2.10
Lowpass FIR filter designed with Blackman window ($M = 61$).

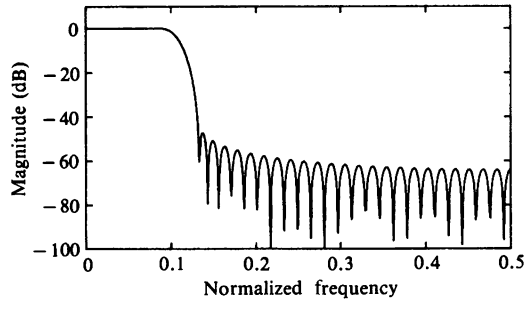


Figure 10.2.11
Lowpass FIR filter designed with $\alpha = 4$ Kaiser window ($M = 61$).